Subject Code	Subject Name	Category	L	T	P	S	Credits	Inst.Hours	Marks		
									CIA	External	Total
23BCE3S2	**Multimedia Systems**	(SEC-V)	2	-	-	-	2	2	25	75	100

	Learning Objectives		
LO1	Understand the definition of Multimedia		
LO2	To study about the Image File Formats, Sounds Audio File Formats		
LO3	Understand the concepts of Animation and Digital Video Containers		
LO4	To study about the Stage of Multimedia Project		
LO5	UnderstandtheconceptofOwnershipofContentCreatedforProjectAcquiringTalent		
	Contents	**No. of Hours**	**Course Objective**
UNITI	Multimedia Definition-Use Of Multimedia-Delivering Multimedia-Text: About Fonts and Faces - Using Text in Multimedia -Computers and Text Font Editing and Design Tools-Hypermedia and Hypertext.	6	
UNITII	Images: Plan Approach-Organize Tools-Configure Computer Workspace -Making Still Images -Color - Image File Formats. Sound: The Power of Sound – Digital Audio-Midi Audio-Midivs.	6	
UNITIII	Digital Audio-Multimedia System Sounds Audio File Formats-Vaughan's Law of Multimedia Minimums- Adding Sound to Multimedia Project.	6	
UNITIV	Animation: The Power of Motion-Principles of Animation-Animation by Computer - Making Animations that Work. Video: Using Video - Working with Video and Displays-Digital Video Containers-Obtaining Video Clips-Shooting and Editing Video.	6	
UNITV	Making Multimedia: The Stage of Multimedia Project - The Intangible Needs -The Hardware Needs - The Software Needs - An Authoring Systems Needs-Multimedia Production Team.	6	
	Total	**30**	

UNIT I

Multimedia Definition

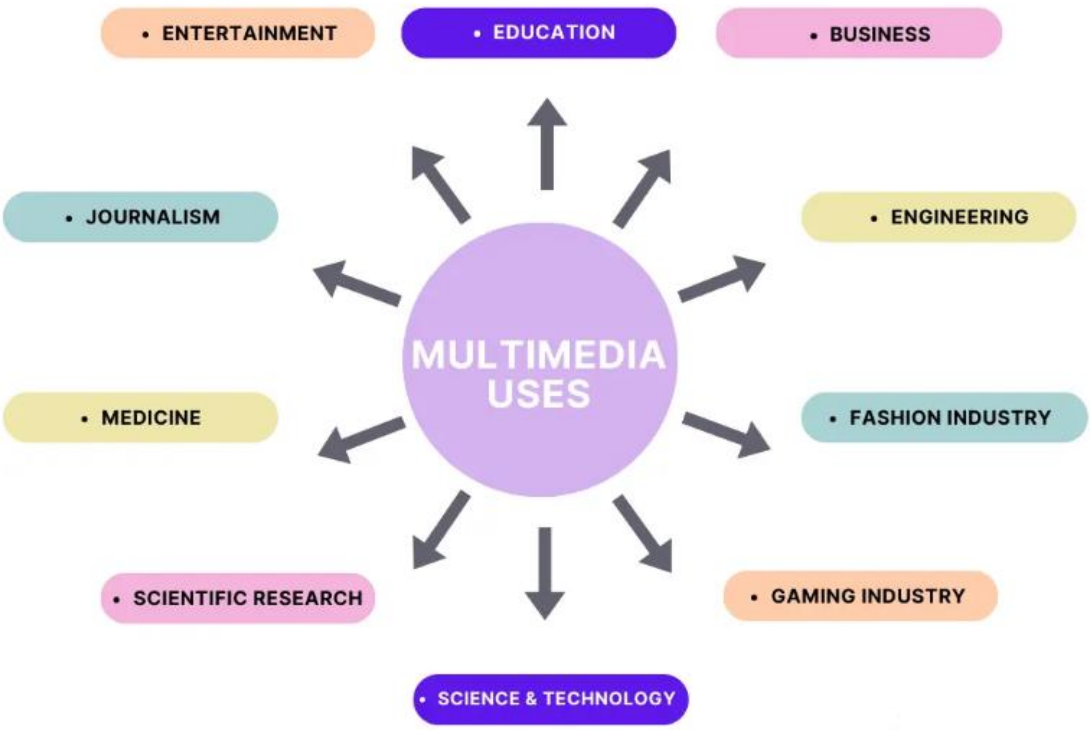
- **Multimedia** is the integration of multiple forms of media — such as **text, images, audio, video, graphics, and animation** — into a single interactive application.
- It combines different content forms to **communicate information more effectively**.
- **Example:** An educational CD or website that includes **text explanations, images, and narration** is a multimedia application.

Use of Multimedia

- **Multimedia** is the combination of **text, graphics, audio, video, and animation** to present information in an engaging and interactive way.

- It helps in **communicating ideas more effectively**, capturing attention, and improving understanding.

Major Uses of Multimedia



1. Education

- Multimedia is widely used in **teaching and learning**.
- It makes learning **interactive, interesting, and easier to understand**.
- Examples:
    - E-learning platforms (Byju's, Coursera)
    - Digital classrooms and simulation-based training

Benefits:

- Increases student engagement
- Supports visual and auditory learning
- Enhances retention of concepts

2. Entertainment

- The **entertainment industry** is one of the biggest users of multimedia.
- Used in **movies, video games, animations, music, and special effects**.
- Examples:
    - 3D animation films
    - Virtual reality gaming

Benefits:

- Creates realistic experiences
- Provides interactive enjoyment

3. Business and Advertising

- Multimedia is used for **product promotion, advertising, and corporate presentations**.
- Used in **digital marketing**, **training videos**, and **interactive company profiles**.

Examples:

- Product demos on websites
- Animated advertisements
- Training and onboarding materials

Benefits:

- Attracts customers effectively
- Simplifies complex business concepts

4. Healthcare

- Used for **medical training**, **virtual surgery simulations**, and **patient education**.
- Doctors use multimedia visuals to explain health conditions and treatments.

Examples:

- 3D anatomy animations
- Health awareness videos

Benefits:

- Improves understanding of medical procedures
- Aids in training healthcare professionals

5. Engineering and Scientific Research

- Multimedia tools help visualize **complex processes and data**.
- Used in **computer simulations**, **CAD (Computer-Aided Design)**, and **data visualization**.

Examples:

- Simulation of machines and systems
- 3D models for design and analysis

Benefits:

- Enhances accuracy and experimentation
- Saves time and cost in design

6. Public Services and Communication

- Used in **information kiosks**, **public awareness campaigns**, and **government websites**.
- Helps in delivering **information clearly and visually** to the public.

Examples:

- Traffic awareness videos
- Digital museum displays

Benefits:

- Reaches a wide audience
- Promotes social awareness

Multimedia plays a **vital role** in today's digital world by making communication more **effective, interactive, and engaging**. Its use spans across **education, business, entertainment, healthcare, and public services**, proving it to be an essential part of modern communication and learning.

Delivering Multimedia

Delivering multimedia means the **method or medium through which multimedia content (text, images, audio, video, animation, etc.) reaches the end user**. The delivery method affects how users **access, interact with, and experience** multimedia applications.

1. Traditional Delivery Methods

a) CD-ROM and DVD

- Earlier multimedia content was distributed through **CD-ROMs and DVDs**.
- Used for **educational software, games, and encyclopedias**.
- Content was **stored offline** and could be accessed without the internet.

Advantages:

- No internet required
- High-quality storage of data

Disadvantages:

- Limited storage space
- Difficult to update content

b) Television and Broadcast Media

- Multimedia is also delivered through **TV channels and satellite broadcasting**.
- Used for **news, advertisements, educational shows, and documentaries**.

Example:

- Educational programs on Discovery Channel or National Geographic.

2. Modern (Digital) Delivery Methods

a) Web-based Multimedia

- The most common delivery platform today.
- Multimedia content is delivered through **websites, social media, and streaming platforms**.
- Examples: **YouTube, Netflix, e-learning portals, interactive websites**.

Features:

- Easily accessible
- Supports audio, video, animation, and text
- Allows interactivity

b) Mobile Multimedia

- Mobile phones and tablets are major platforms for multimedia delivery.
- Examples: **Educational apps, games, social media posts, AR/VR experiences**.

Advantages:

- Portable and user-friendly
- Provides instant access to multimedia content

c) Streaming Media

- Audio and video files are delivered in **real time** over the internet without full downloading.
- Examples: **Spotify, Zoom, YouTube Live, Netflix**.

Benefits:

- Saves time and storage space
- Allows live broadcasts and real-time learning

d) Cloud-based Delivery

- Multimedia content is stored on **cloud servers** and accessed remotely.
- Examples: **Google Drive, Dropbox, OneDrive**.

Advantages:

- Easy sharing and collaboration
- Access from any device

3. Factors Affecting Multimedia Delivery

1. **File Size:** Large multimedia files need compression to reduce load time.
2. **Bandwidth:** Determines speed and quality of online multimedia.
3. **Hardware/Software Requirements:** Users need suitable devices and players.
4. **Interactivity Level:** Some systems require user input or navigation.
5. **Format Compatibility:** Files must support playback on different platforms.

Delivering multimedia effectively ensures that **users experience high-quality, accessible, and engaging content**. With advancements in **internet and mobile technologies**, multimedia delivery has become **faster, more interactive, and globally available**, enabling learning, entertainment, and communication anywhere, anytime.

Text: About Fonts and Faces

- Text is the **most basic and essential element** in multimedia.
- It provides information, instructions, and communication through written words.
- The **appearance of text** — such as its style, size, and color — plays a vital role in how effectively information is presented and understood.

1. What are Fonts and Faces?

a) Typeface

- A **typeface** is a **family of characters** (letters, numbers, symbols, punctuation marks) that share a common design.
- Examples: **Arial, Times New Roman, Verdana, Calibri**.

Each typeface has different **styles** (bold, italic, underline, etc.) that change its appearance but retain the same basic design.

b) Font

- A **font** is a specific **style, weight, and size** of a typeface.
- Example:
    - Typeface → *Times New Roman*
    - Fonts → *Times New Roman Bold, Times New Roman Italic, Times New Roman 12 pt*

In short:

- *Typeface* = design family

- *Font* = specific member of that family

2. Font Attributes

Fonts have several characteristics that affect readability and design quality:

1. **Size:** Measured in points (pt). Example: 12 pt, 14 pt.
2. **Style:** Regular, Bold, Italic, Underline, etc.
3. **Color:** Used to highlight or emphasize text.
4. **Alignment:** Left, right, center, or justified.
5. **Spacing:** Refers to the space between characters (kerning) and lines (leading).

3. Types of Typefaces

1. **Serif Fonts:**
    - Have small decorative strokes (serifs) at the ends of letters.
    - Example: **Times New Roman, Georgia**.
    - Best for **printed documents** as they guide the reader's eye smoothly.
2. **Sans-Serif Fonts:**
    - "Sans" means "without" — these fonts don't have serifs.
    - Example: **Arial, Helvetica, Calibri**.
    - Best for **digital screens** due to clear and modern look.
3. **Script Fonts:**
    - Look like handwriting or calligraphy.
    - Example: **Brush Script, Lucida Handwriting**.
    - Used for **invitations, logos, or headings**.
4. **Decorative / Display Fonts:**
    - Artistic and designed to attract attention.
    - Example: **Chiller, Comic Sans, Jokerman**.
    - Used for **titles, banners, or posters**, not for large text blocks.

4. Importance of Fonts in Multimedia

- Fonts create the **visual personality** of a project.
- Help in **setting mood and tone** (formal, playful, professional, etc.).
- Influence **readability and user experience**.
- Consistent and appropriate fonts improve **communication clarity**.

5. Guidelines for Using Fonts in Multimedia

1. Use **simple and readable fonts** for body text.
2. Use **contrasting fonts** for headings and subheadings.
3. Avoid using **too many font styles** in one project.
4. Maintain **color contrast** between text and background.
5. Ensure text size and styles are **consistent** throughout.

Fonts and faces play a **crucial role** in multimedia design. They affect not only the **readability** but also the **emotional appeal** and **aesthetic quality** of a presentation. Choosing the right typeface and font helps deliver the message **clearly, attractively, and effectively**.

Using Text in Multimedia

- Text is one of the **most important elements** of multimedia.
- It is the **primary source of information** and is used to communicate ideas, messages, and instructions.
- In multimedia applications, text works together with images, sound, video, and animation to create an **effective and engaging presentation**.

1. Importance of Text in Multimedia

- Text provides **clarity and structure** to multimedia content.
- It helps users **navigate, read, and understand** the information.
- Text can act as a **supporting element** to visuals and sounds.
- It allows for **interaction** through hyperlinks, menus, and buttons.

2. Ways Text is used in Multimedia

1. **Titles and Headings**
    - Used to introduce topics, sections, or screens.
    - Example: *"Welcome to the Learning Module"*
2. **Content or Body Text**
    - Used to display detailed information, explanations, or instructions.
    - Example: *Tutorial descriptions or lesson texts.*
3. **Menus and Navigation Text**
    - Helps users move through multimedia content easily.
    - Example: *Home, About, Next, Exit buttons.*
4. **Labels and Captions**
    - Used to describe images, videos, or animations.
    - Example: *Figure 1: Solar Energy Panel.*
5. **Interactive Text (Hypertext)**
    - Contains clickable links that take users to other pages or media.
    - Example: *Click here to learn more.*
6. **Credits and Subtitles**
    - Used in videos or presentations to display names, roles, or translations.

3. Guidelines for Using Text Effectively

1. **Readability:**
    - Use clear and simple fonts like *Arial* or *Verdana*.
    - Avoid using too many font types in one screen.
2. **Font Size and Style:**
    - Use large, bold fonts for headings.
    - Use smaller but readable fonts for body text.
3. **Color and Contrast:**
    - Maintain proper contrast between text and background.
    - Example: *Black text on white background.*
4. **Consistency:**
    - Keep font style, size, and color uniform throughout the project.
5. **Text Placement:**
    - Place text near related images or visuals for easy understanding.

6. **Animation Effects:**
    - Use text animations (fade-in, slide, zoom) to add visual appeal — but don't overuse them.

4. Role of Text in Different Multimedia Applications

Application Area	Role of Text	Example
Education	Explains lessons and instructions	E-learning modules
Business	Describes products, adds interactivity	Company presentations
Websites	Provides content and navigation	Menus, blogs, hyperlinks
Entertainment	Subtitles, credits, lyrics	Movies, music videos
Games	Instructions, scores, dialogues	Game interface text

5. Tools for Using Text in Multimedia

- **Microsoft PowerPoint** – For text-based slides and animations
- **Adobe Photoshop / Illustrator** – For creating stylish text graphics
- **Flash / Animate** – For animated text effects
- **HTML / CSS** – For web-based text formatting
- **Video Editors (Premiere Pro, Filmora)** – For subtitles and captions

Text is a **powerful communication tool** in multimedia. It adds **meaning, guidance, and structure** to multimedia presentations. When combined with other elements like images, audio, and video, text enhances **understanding, interactivity, and user experience**. Therefore, using text **wisely and attractively** is essential for creating effective multimedia applications.

Computers and Text Font Editing and Design Tools

1. Computers and Text

- Computers play a major role in **creating; storing, editing, and displaying text** in multimedia projects.
- Text in multimedia applications is created and controlled using computer software and digital technologies.

a) Representation of Text

- Computers represent text using **character encoding systems** such as:
    - **ASCII (American Standard Code for Information Interchange)**
    - **Unicode** (supports multiple languages and special symbols)
- Each character (letter, number, or symbol) is represented by a unique binary code (0s and 1s).

b) Role of Computers

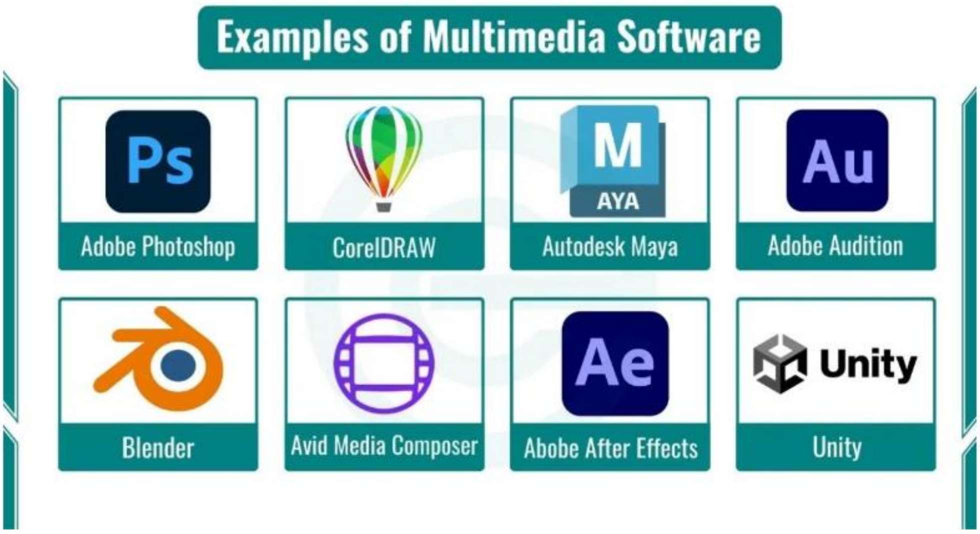
- Computers are used to **input, format, design, animate, and combine text** with other multimedia elements such as images, audio, and video.
- They make it easy to **change font size, color, style, and alignment** instantly.
- With the help of multimedia software, computers enable **interactive and animated text effects**.

Example:
Creating animated titles in PowerPoint or stylized text in Photoshop.

2. Font Editing and Design Tools

- Font editing and design tools are software programs used to **create, modify, and style fonts** for multimedia projects.
- They help designers give text a **unique look and feel** suitable for the project's theme.

3. Common Font Editing and Design Tools



a) Adobe Illustrator

- Professional vector-based design software.
- Used for **creating logos, titles, and custom typography**.
- Allows precise control over shapes, curves, and letter spacing.

b) CorelDRAW

- Used to **design artistic and decorative text** for posters, banners, and advertisements.
- Offers features like text-to-path, shadow effects, and 3D text styling.

c) Adobe Photoshop

- Used to **add special effects** to text such as glow, shadow, gradient, and texture.

- Helps integrate text with images and backgrounds.
- Ideal for web graphics and digital artwork.

d) FontForge

- A **free and open-source font editor**.
- Allows users to **create new font styles** or modify existing ones (TrueType, OpenType).
- Used by designers to make customized fonts.

e) FontLab Studio / Glyphs

- Professional tools for **font design and font engineering**.
- Used for creating commercial fonts used in branding and publishing.

f) Microsoft Word / PowerPoint

- Commonly used for **basic text formatting** and **presentation designs**.
- Supports font selection, text color, size adjustment, and simple effects.

4. Importance of Font Editing and Design in Multimedia

1. **Visual Appeal:**
   - Enhances the look of presentations, websites, and animations.
2. **Brand Identity:**
   - Unique fonts help represent a brand's style and message.
3. **Readability and Clarity:**
   - Proper design ensures that text is clear and easy to read.
4. **Consistency:**
   - Font tools help maintain uniformity across all screens and media.
5. **Creativity:**
   - Allows designers to experiment with artistic styles to make text attractive.

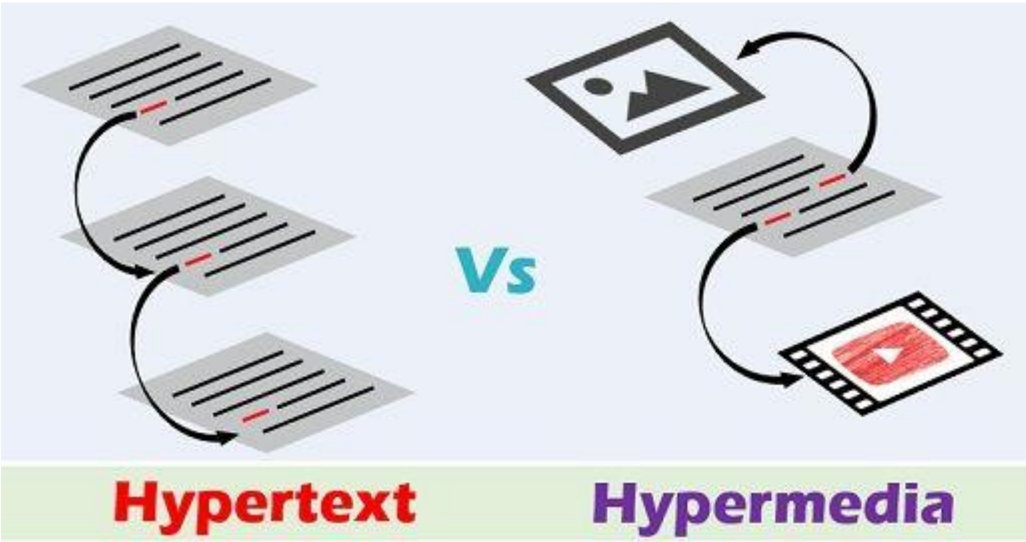
5. Applications in Multimedia Projects

- Designing **titles and logos** for videos or games.
- Creating **animated text** for intros and transitions.
- Styling **content and captions** for presentations and websites.
- Making **digital advertisements** with special font effects.

Computers make it possible to **create, edit, and display text dynamically** in multimedia. With the help of **font editing and design tools**, designers can turn ordinary text into **visually appealing and expressive elements**. Choosing and designing the right font helps ensure the multimedia project is **attractive, readable, and impactful**.

Hypermedia and Hypertext

In multimedia systems, **hypertext** and **hypermedia** are techniques that allow users to **navigate information non-linearly** — meaning users can jump from one piece of information to another using **links** (called *hyperlinks*). These concepts make multimedia **interactive and user-friendly**.



1. Hypertext

- Hypertext refers to **text that contains links (hyperlinks)** to other pieces of text or documents.
- It allows users to **navigate between related topics or documents** easily by clicking on these links.

Example:

When reading an online article, if you click on a word or phrase that takes you to another related article or page — that is **hypertext**.

Key Features:

- **Non-linear navigation:** Users can jump to any section they wish.
- **Text-based linking:** Mainly involves words and phrases.
- **Interactive:** Readers can choose their own reading path.

Example Applications:

- Wikipedia pages (links between articles)
- Online textbooks and documentation
- HTML web pages

2.Hypermedia

Hypermedia is an **extension of hypertext** that includes **multiple forms of media** — such as text, images, audio, video, graphics, and animations — all interconnected using hyperlinks.

It allows users to **browse and interact** with various types of multimedia elements in a single application or webpage.

Example:

A learning website where clicking on a word opens a related video or image explanation is an example of **hypermedia**.

Components of Hypermedia:

- **Text** – the basic medium of information
- **Images** – diagrams, pictures, illustrations
- **Audio** – sound clips, voice explanations
- **Video** – motion visuals, tutorials
- **Animation** – interactive demonstrations

Example Applications:

- Educational CDs or e-learning modules
- Multimedia encyclopedias
- Interactive museum kiosks
- Web-based training programs

Difference between Hypertext and Hypermedia:

S.No.	HYPERTEXT	HYPERMEDIA
01.	Hypertext refers to the system of managing the information related to the plain text.	Hypermedia refers to connecting hypertext with other media such as graphics, sounds, animations.
02.	Hypertext involves only text.	It involves graphics, image, video, audio etc.
03.	Here only text becomes the part of link.	It is enhanced version of hypertext here along with text other multimedia also becomes the part of link.
04.	Hypertext is the part of hypermedia means it comes under hypermedia.	Hypermedia is the superior entity which is the advanced version of hypertext.
05.	It allows the user to navigate through text in a non linear way.	It includes multimedia to improve the multimedia experience.

06.	It simply allows users to move from one document to another with a single click on the hypertext or goto links.	It extends the ability of hypertext and allows user to click text or any other multimedia to move one to another page.
07.	Hypertext attracts to the user to move around the document as well as to move from one page of document to another.	Hypermedia attracts the user more than the hypertext as it gives more flexibility of movement.
08.	It provides a less user experience to the users than the hypermedia.	It provides a better user experience to the users than the hypertext.

Advantages of Hypertext and Hypermedia

1. **Interactive Learning:**

   Users can explore topics in any order, enhancing understanding.

2. **Easy Navigation:**

   Hyperlinks make it simple to move between related information.

3. **Rich Presentation:**

   Combining text, video, and sound makes content engaging.

4. **Efficient Information Access:**

   Quick access to additional references or resources.

5. **User Control:**

   The learner controls what to read, view, or explore next.

UNIT II

Images

- An **image** is a **visual representation** of a person, object, scene, or idea used to communicate information or enhance the appearance of a multimedia project.
- Images can be **still (static)** like photographs, drawings, charts, and diagrams, or **moving** like animations.

Types of Images:

1. **Still Images** – Photos, illustrations, logos, etc.
2. **Dynamic Images** – Animations or motion graphics.

Uses of Images in Multimedia:

- To make presentations attractive.
- To explain ideas clearly (e.g., diagrams in e-learning).
- To create brand identity (e.g., logos, posters).
- To provide background or decoration in web pages and apps.

Examples of Image Formats:

- **JPEG (.jpg):** Common for photos.
- **PNG (.png):** Supports transparency.
- **GIF (.gif):** Used for short animations.
- **SVG (.svg):** Used for scalable vector graphics.

Plan Approach

- Creating images for multimedia projects requires **planning** to ensure that visuals communicate the intended message effectively.
- A well-planned approach improves quality, saves time, and maintains consistency.

Steps in Planning:

1. **Define the Purpose:**
    - Identify why you need the image — for learning, advertising, entertainment, etc.
    - Example: Educational projects use labeled diagrams, while advertisements use eye-catching visuals.
2. **Identify the Target Audience:**
    - Choose image styles, colors, and complexity according to the audience's age and interest.
    - Example: Bright colors for children, professional tones for business users.
3. **Select the Image Type:**
    - Choose whether the image should be a **photograph**, **illustration**, **icon**, or **animation frame**.
4. **Choose the Image Source:**

- Images can be **captured** (camera), **scanned**, **drawn**, or **downloaded** from copyright-free sources.

5. **Plan for Consistency:**
    - Use a uniform style and color theme across all visuals to create a professional look.
6. **Optimize File Format and Size:**
    - Choose suitable formats (JPEG, PNG, GIF) depending on the platform (web, print, app).

Example:
A school e-learning project might use PNG diagrams with transparent backgrounds to blend well with colored slides.

Organize Tools

- To make or edit images, designers use various software tools.
- Selecting and organizing tools properly improves productivity and output quality.

Common Image Creation and Editing Tools:

- **Adobe Photoshop:** Professional tool for photo editing, retouching, and design.
- **CorelDRAW:** Ideal for vector-based illustrations, logos, and posters.
- **GIMP:** Free and open-source alternative to Photoshop.
- **Canva:** Easy-to-use online tool for quick web and social media graphics.
- **Illustrator:** Used for creating scalable vector art and branding materials.

Organizing Your Workspace:

- Create separate folders for **raw images**, **edited versions**, and **final outputs**.
- Maintain **consistent naming** (e.g., "logo_final_v2.png").
- Backup files regularly to prevent data loss.

Configure Computer Workspace

An efficient computer setup helps in working with high-quality images without performance issues.

Hardware Requirements:

- **Monitor:** High-resolution screen with color accuracy.
- **Processor (CPU):** Multi-core for faster rendering.
- **RAM:** Minimum 8GB (16GB preferred for large images).
- **Graphics Card:** For smooth editing and 3D rendering.
- **Storage:** SSD for faster file loading and saving.

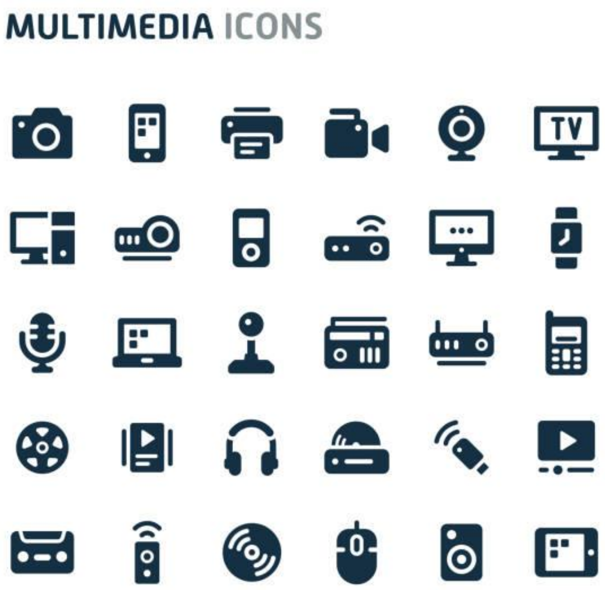
Software and Settings:

- Install **photo editing applications** and **color management tools**.
- Calibrate the monitor for **accurate color display**.

- Set up a **comfortable workspace** (desk lighting, posture, eye-level screen).

Making Still Images

Still images are **non-moving visual representations**, such as photos, illustrations, or graphics that communicate ideas effectively.



Methods of Creating Still Images:

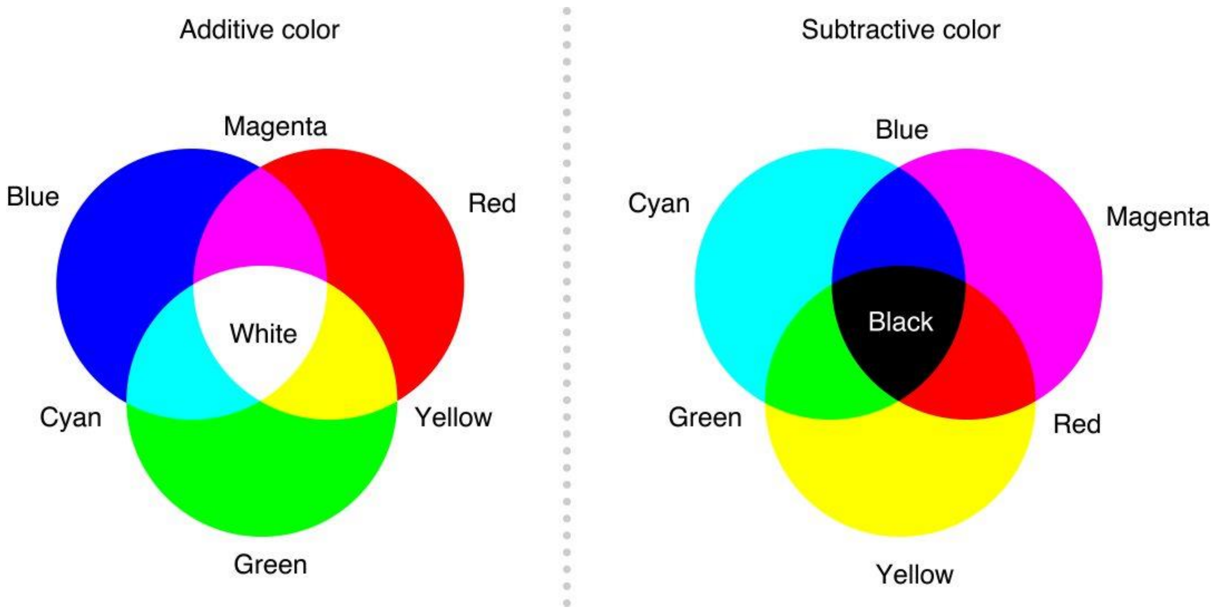
1. **Photography:**
   - Capturing real-world scenes using digital cameras or smartphones.
   - Example: Product photos for e-commerce websites.
2. **Scanning:**
   - Converting printed photos, drawings, or documents into digital format using a scanner.
3. **Drawing or Painting Digitally:**
   - Creating illustrations using software like Photoshop or Procreate.
4. **Image Editing:**
   - Modifying or enhancing images using tools (cropping, brightness, filters).
5. **Compositing:**
   - Combining multiple images into one for creative designs (posters, collages).

Uses of Still Images:

- To explain concepts in e-learning (diagrams, charts)
- To enhance user interfaces (icons, buttons)
- For advertising and branding (logos, banners)

Color

Color plays a **powerful role** in visual communication. It attracts attention, creates emotion, and improves recognition.



Types of Color Models:

- **RGB (Red, Green, Blue):**
  Used for digital screens. All colors are made by mixing red, green, and blue light.
  Example: Computer monitors, smartphones.
- **CMYK (Cyan, Magenta, Yellow, Black):**
  Used for printing. Based on pigment mixing.
- **HSB (Hue, Saturation, Brightness):**
  Used for adjusting tones in image editing tools.

Color Depth:

- Determines how many colors an image can display.
  Example:
  - 8-bit → 256 colors
  - 24-bit → 16 million colors

Importance of Color:

- Creates mood and emotion (warm colors = energy; cool colors = calm).
- Emphasizes key information.
- Maintains brand consistency.

Image File Formats

Images can be stored in different file types depending on their purpose and level of compression.

Format	Full Form	Type	Features / Uses
JPEG (.jpg)	Joint Photographic Experts Group	Lossy	Small file size, ideal for photos and web use
PNG (.png)	Portable Network Graphics	Lossless	Supports transparency, high quality

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Alagappa University.

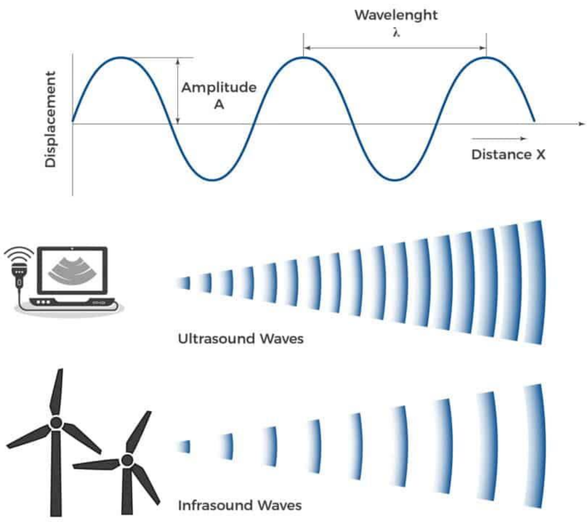
GIF (.gif)	Graphics Interchange Format	Lossless	Supports animation, limited to 256 colors
TIFF (.tif)	Tagged Image File Format	Lossless	Used in professional printing
BMP (.bmp)	Bitmap	Uncompressed	Large file, used in Windows apps
SVG (.svg)	Scalable Vector Graphics	Vector	Resolution-independent, used for icons/logos
WebP / HEIF	Modern Web Formats	Compressed	Web-optimized, smaller yet high quality

Sound:

Sound is an **audio element** used to convey information, create mood, and make multimedia content more engaging and realistic. It can include **speech, music, sound effects, and background audio**.

Types of Sound:

1. **Speech / Narration:** Used in tutorials or guides.
2. **Music:** Sets mood and emotion.
3. **Sound Effects:** Adds realism (click, beep, thunder, etc.).
4. **Ambient Sound:** Background environment noise.



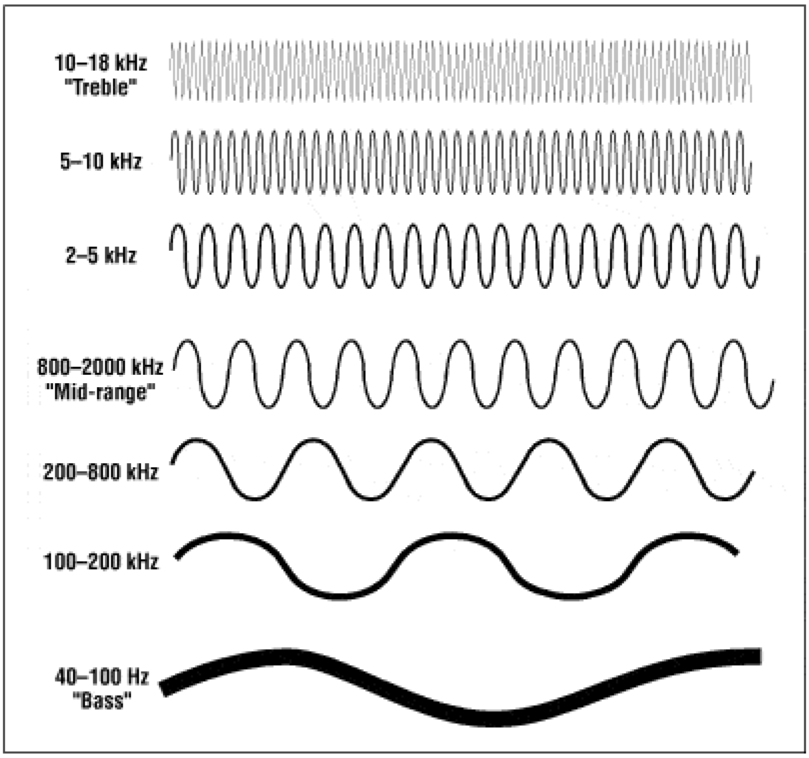
Forms of Digital Sound:

- **Digital Audio:** Actual recorded sound stored in digital form (e.g., MP3, WAV).

- **MIDI (Musical Instrument Digital Interface):** Stores musical instructions, not actual sound.

Uses of Sound in Multimedia:

- To catch attention and improve user experience.
- To provide audio feedback in applications.
- To support storytelling in animations and videos.
- To make e-learning more interactive and memorable.



The Power of Sound

Sound gives **life, emotion, and depth** to multimedia projects.
It helps communicate messages more effectively and enhances user experience.

Uses of Sound:

- **Narration / Voice-over:** Guides users or explains visuals.
- **Background Music:** Creates mood and atmosphere.
- **Sound Effects:** Adds realism (clicks, alerts, explosions, etc.).
- **Interactive Audio:** Responds to user actions in apps and games.

Example:

In a children’s learning app, background music and voice instructions keep users engaged and attentive.

Digital Audio

Digital audio converts real-world (analog) sounds into binary data that computers can process.

Steps:

1. **Recording:** Sound waves are captured using a microphone.
2. **Analog-to-Digital Conversion (ADC):** Converts sound into samples.
3. **Editing:** Adjusting volume, pitch, and clarity using software.
4. **Playback:** Sound is converted back to analog form for speakers.

Characteristics of Digital Audio:

- **Sampling Rate:**
  - The number of samples taken per second (Hz).
  - CD Quality = 44.1 kHz (44,100 samples/sec).
- **Bit Depth:**
  - Determines precision of sound (8-bit, 16-bit, 24-bit).
- **Channels:**
  - **Mono:** Single channel.
  - **Stereo:** Two channels for spatial sound.

Common Audio File Formats:

Format	Full Form	Features / Use
WAV (.wav)	Waveform Audio File	Uncompressed, high quality
MP3 (.mp3)	MPEG Audio Layer 3	Compressed, small size
AAC (.aac)	Advanced Audio Coding	Better than MP3, used in iTunes
OGG (.ogg)	Open Source Audio Format	Free and good for web
FLAC (.flac)	Free Lossless Audio Codec	High-quality, lossless

Midi Audio

MIDI (Musical Instrument Digital Interface) is a protocol that stores **musical instructions** rather than actual sound waves.

Features:

- Very **small file size**.
- Allows **editing** of tempo, pitch, and instrument.
- Played using **software synthesizers** or **digital keyboards**.

Advantages:

- Easy to modify.

- Requires less memory.
- Perfect for background scores in games or animations.

Example:

A MIDI file might tell a computer to play a melody using virtual piano sounds.

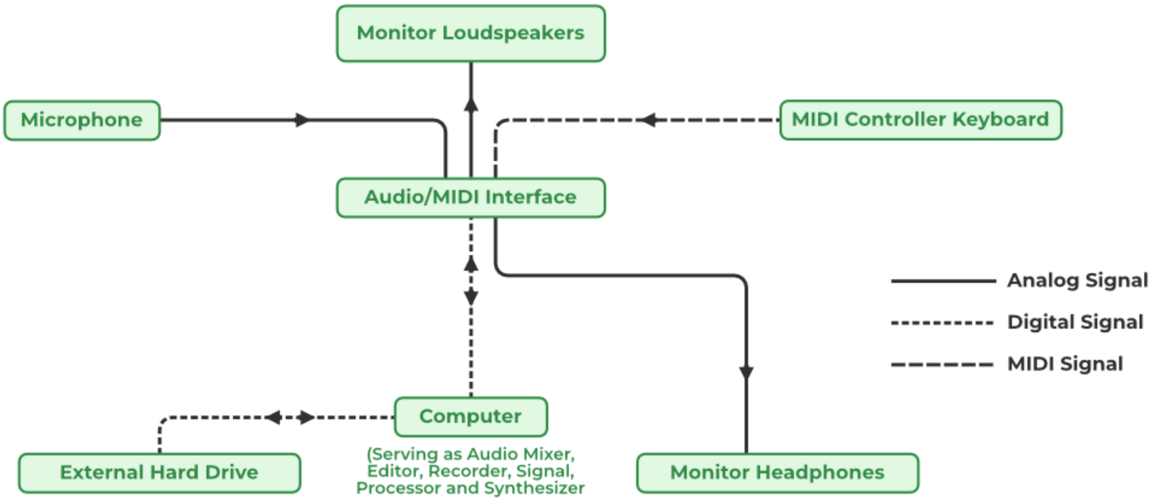
Midivs.

Aspect	MIDI Audio	Digital Audio
Nature	Instructions for notes	Actual recorded sound
File Size	Small	Large
Quality	Depends on synthesizer	Consistent, high
Editing	Easy (change instruments)	Limited
Use	Music composition, games	Voice, sound effects

UNIT III

Digital Audio

- ✓ **Definition**: Digital audio is sound stored in a computer-readable format. It is created by converting natural sound waves into **binary numbers (0s and 1s)** through **sampling** and **quantization**.
- ✓ Binary data is used to represent sound in digital audio. It entails transforming analog sound waves into a digital version that can be stored, handled and transferred using digital technology.



Key Concepts:

- **Sampling Rate** – Number of sound samples taken per second (e.g., CD quality = **44.1 kHz**).
- **Bit Depth** – Number of bits used per sample (higher bit depth = better quality).
- **Channels** – Mono (1 channel), Stereo (2 channels), Surround (5.1, 7.1).

Example: A **music track in MP3 format** is stored at 128 kbps compression, while a **CD-quality song in WAV format** is uncompressed at 44.1 kHz, 16-bit.

Multimedia System Sounds

Sounds used in multimedia projects include:

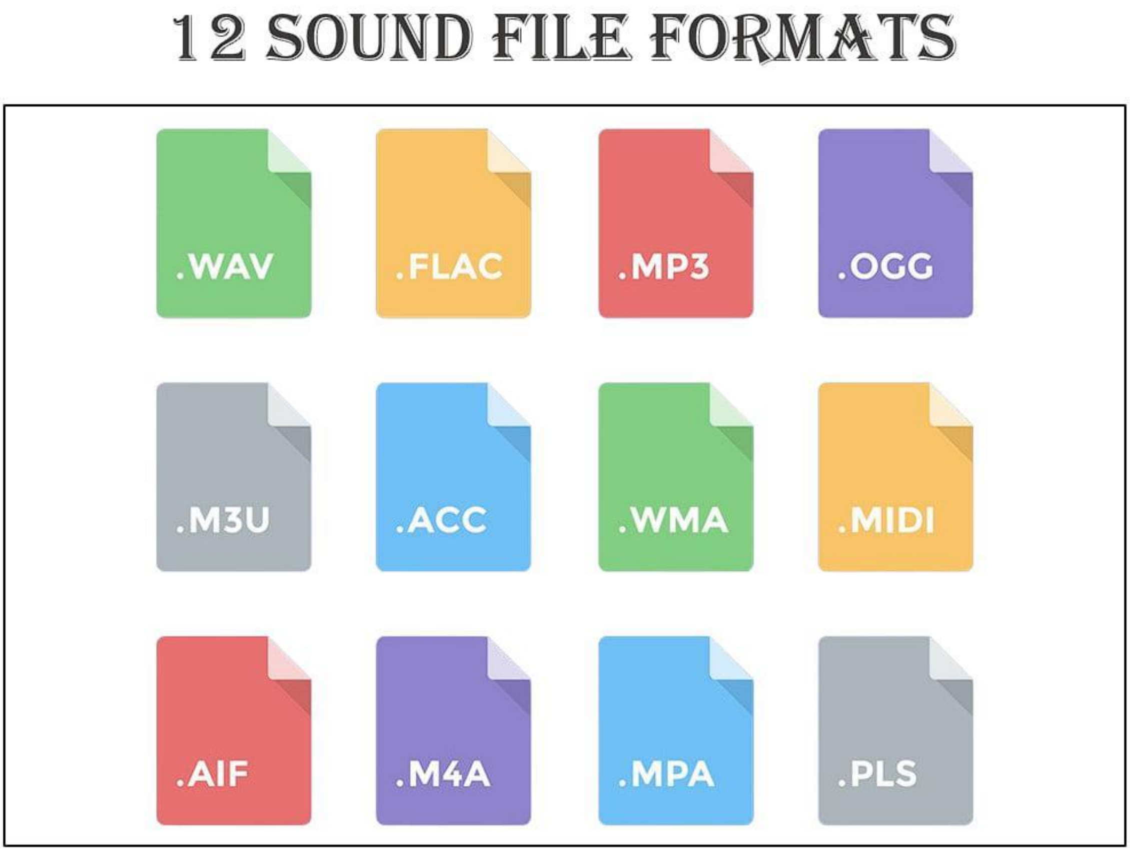
- **Voice narration** – For explanations in e-learning apps.
- **Music** – Background music in movies or games.
- **Sound effects (SFX)** – Click sounds in apps, explosions in games.
- **Ambient sounds** – Nature sounds in meditation apps.

Example: In a **children's learning app**, narration explains alphabets, background music keeps it lively, and sound effects play when clicking objects.

Audio File Formats

An audio file format is a file format for storing digital audio data on a computer system. The bit layout of the audio data (excluding metadata) is called the audio coding format and can be uncompressed, or compressed to reduce the file size, often using lossy compression.

Common formats for storing audio:



- **WAV (.wav)** – Uncompressed, high quality, large size.
  - *Example*: Used in professional studios for editing.
- **MP3 (.mp3)** – Compressed, small size, widely used.
  - *Example*: Songs on mobile phones.
- **AAC (.aac)** – Advanced compression, better than MP3.
  - *Example*: iTunes and YouTube audio.
- **WMA (.wma)** – Windows Media Audio.
- **MIDI (.mid)** – Stores musical notes, not real sound.
  - *Example*: Background tunes in old mobile phones.

- **OGG (.ogg)** – Open-source compressed format.

4. Vaughan's Law of Multimedia Minimums

- Proposed by **Tay Vaughan** (a multimedia expert).
- States that **only the minimum elements required to communicate the message should be used**.
- Too much sound, text, or animation can **distract and confuse** the user.

Example: An **educational video** should have clear narration and visuals — not loud background music that distracts learners.

Multimedia is the integration of various media types, like text, graphics, audio, video, and animation, to create a richer and more engaging experience. Think of it as using more than just words to communicate an idea.

Here are some common examples:

- **Websites:** Modern websites often combine text with images, embedded videos, and interactive elements.
- **Video Games:** These are a classic example, blending graphics, animation, sound effects, music, and text (for instructions or dialogue) to create an interactive world.
- **Presentations (e.g., PowerPoint):** These presentations use text, images, charts, and sometimes audio or video clips to convey information.
- **Online Learning Platforms:** They use text, videos, interactive quizzes, and graphics to help students learn.
- **Films and Television:** These combine moving images, sound, music, and text (like subtitles) for storytelling and entertainment.
- **Social Media:** Platforms like Instagram, TikTok, and Facebook heavily rely on multimedia, allowing users to share photos, videos, audio clips, and text.
- **Interactive Kiosks:** Found in museums or public spaces, these use touchscreens with graphics, text, and video to provide information.

Vaughan's Law of Multimedia Minimums

This law, coined by Tay Vaughan, suggests that **there's an acceptable minimum level of quality that satisfies an audience**, even if it's not the absolute best technology or effort could produce.

Example: Imagine you're creating a training video for employees.

- **Scenario 1 (Beyond the Minimum):** You spend a fortune on high-definition cameras, professional voice actors, and complex animations. While it looks and sounds amazing, the core training information might be lost in the polish, or employees might find it overly flashy and distracting.
- **Scenario 2 (Meeting the Minimum):** You use clear visuals, a well-recorded voiceover explaining the process, and maybe a few simple animations to highlight key steps. The information is easy to understand, the audio is clear, and the video quality is good enough. According to Vaughan's Law, this "good enough" approach, if it effectively conveys the training, is sufficient and may be more practical than overspending on bells and whistles that don't significantly improve the learning outcome.

In essence, the law encourages focusing on **functional adequacy** and user needs rather than chasing the absolute cutting edge if it doesn't add tangible value.

5. Adding Sound to Multimedia Project

Steps:

1. **Collect / Record** sound
   - Microphone, digital recorder, or royalty-free sound libraries.
   - *Example*: Record a voice-over for a tutorial video.
2. **Edit** the sound
   - Software: Audacity, Adobe Audition, GarageBand.
   - *Example*: Remove noise from narration.
3. **Compress & Save** in suitable format
   - WAV for quality, MP3 for small size.
   - *Example*: Save an app’s background music as MP3 to reduce size.
4. **Integrate** sound into multimedia project
   - Insert into PowerPoint, Video Editor, Unity, or e-learning software.
   - *Example*: Syncing background score with animation in a short film.
5. **Synchronize** with visuals
   - Timing is crucial so sound matches the action.
   - *Example*: Gunfire sound in a game must match the bullet animation.

- **Digital audio** stores sound in binary.
- Multimedia systems use narration, music, sound effects, and ambient sounds.
- Formats include **WAV, MP3, AAC, MIDI**.
- **Vaughan’s Law**: Use minimum sound needed to deliver the message.

- Adding sound involves **recording, editing, saving, integrating, and synchronizing**.

UNIT IV

Animation:

- **Animation** is the process of creating the **illusion of movement** by displaying a series of still images (called *frames*) one after another very quickly.
- When these images are shown in rapid sequence (like 24 or 30 frames per second), our eyes see them as a moving picture.

� Examples:

- Cartoon shows like *Tom and Jerry*
- Animated movies like *Toy Story* or *Frozen*
- Moving icons or characters in games
- Educational animations in e-learning videos

� Uses of Animation:

- To make multimedia presentations more interesting
- To explain complex ideas through motion
- In movies, games, and advertisements
- In websites and mobile apps for interactive effects

The Power of Motion

Animation is the **process of creating the illusion of movement** by displaying a series of still images (frames) in rapid succession.

When images are shown quickly (usually 24–30 frames per second), our eyes perceive them as moving.

Importance of Animation:

- Brings **life and movement** to multimedia content.
- Makes learning **interactive and entertaining**.
- Helps in **visual storytelling** and demonstrating complex concepts.
- Used widely in **education, advertising, gaming, movies, and simulations**.

Examples:

- Animated cartoons
- Explainer videos
- 3D architectural walkthroughs
- Motion info graphics

Principles of Animation

- The principles of animation were first defined by Disney animators and are still used today in both 2D and 3D animation.
- They help make animations more **realistic, expressive, and appealing**.

Key Principles:

1. **Squash and Stretch**
    - Gives a sense of weight and flexibility.
    - Example: A bouncing ball squashes on impact and stretches when moving up.
2. **Anticipation**
    - Prepares the audience for an action.
    - Example: A person bends before jumping.
3. **Staging**
    - Presenting the action clearly and drawing attention to the main idea.
4. **Straight Ahead Action and Pose to Pose**
    - **Straight Ahead:** Drawing each frame one by one (natural motion).
    - **Pose to Pose:** Creating key poses first, then filling in between (controlled animation).
5. **Follow Through and Overlapping Action**
    - Parts of a body or object continue to move after the main action has stopped.
6. **Slow In and Slow Out**
    - Objects accelerate and decelerate naturally.
7. **Arcs**
    - Most movements follow curved paths, not straight lines.
8. **Timing**
    - Number of frames used for an action affects its speed and mood.
9. **Exaggeration**
    - Enhancing actions or expressions for dramatic effect.
10. **Secondary Action**
    - Additional actions that support the main motion (like hair bouncing as a person runs).
11. **Appeal**
    - Making characters and visuals attractive and interesting to watch.

Animation by Computer

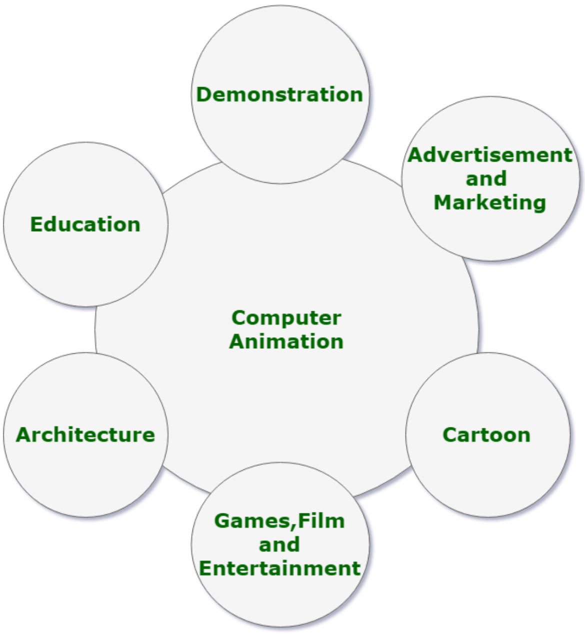
Computer animation involves using **software and digital tools** to create motion graphics, 2D/3D animations, and effects.

Types of Computer Animation:

1. **2D Animation:**
    - Based on flat drawings or vector graphics.
    - Used in cartoons, explainer videos, and web animations.

- Tools: Adobe Animate, Toon Boom, Synfig.

2. **3D Animation:**
    - Uses 3D models, lighting, and rendering for realism.
    - Used in movies, games, and simulations.
    - Tools: Blender, Autodesk Maya, 3ds Max.
3. **Motion Graphics:**
    - Text and graphic-based animation for advertisements or intros.
    - Tools: Adobe After Effects, DaVinci Resolve.
4. **Stop Motion:**
    - Physical objects are photographed frame by frame to simulate motion.
    - Example: Claymation.



Making Animations that Work

To make an effective and professional animation, proper **planning, designing, and testing** are essential.

Steps for Creating Animation:

1. **Concept and Storyboard:**
    - Define the idea and create a visual plan (sequence of scenes).
2. **Design and Modeling:**
    - Create characters, backgrounds, and objects.
3. **Animation:**
    - Apply motion using keyframes, paths, or physics simulation.
4. **Sound and Music:**
    - Add narration, sound effects, and background music to enhance impact.
5. **Rendering and Exporting:**
    - Convert the animation into video format for display.
6. **Testing and Optimization:**

- Ensure smooth playback and adjust timing or effects if needed.

Tips for Effective Animation:

- Keep motion smooth and natural.
- Avoid overuse of effects.
- Synchronize sound properly.
- Focus on the message, not just visuals.

Video:

- **Video** is a sequence of moving images combined with sound.
- It records **real-life motion** using a camera and plays it back on a screen.

▯ Examples:

- Movies and documentaries
- YouTube tutorials
- Recorded lectures or presentations
- News broadcasts

▯ Uses of Video:

- To show real-world events and actions
- For education, entertainment, and communication
- In multimedia projects, advertisements, and social media
- In training and promotional materials

Using Video

Video is the **combination of moving images and sound** used to convey information dynamically. It is an essential part of multimedia for **education, entertainment, marketing, and training**.

Advantages of Using Video:

- Grabs attention and maintains user interest.
- Demonstrates real-world actions and processes.
- Combines visuals and sound for stronger communication.

Applications:

- Online classes and tutorials
- Promotional videos and advertisements
- Documentaries and news
- Social media content

Working with Video and Displays

When using video in multimedia, understanding how it will be **displayed and played** is important.

Video Display Factors:

- **Resolution:** Number of pixels in the frame (e.g., 1920×1080 for Full HD).
- **Frame Rate:** Frames per second (fps) — 24 fps for film, 30 fps for TV.
- **Aspect Ratio:** Ratio of width to height (e.g., 16:9 widescreen).
- **Color Depth:** Number of colors displayed (8-bit, 10-bit, etc.).
- **Compression:** Reducing file size for smooth playback.

Display Devices:

- Computer monitors
- Projectors
- LED / LCD / OLED screens
- Mobile and VR displays

Digital Video Containers

- A **video container** is a file format that stores both **video and audio data** together.
- It may also include **subtitles, chapters, and metadata**.

Common Digital Video Formats:

Format	Full Form	Description
AVI (.avi)	Audio Video Interleave	Older format, large file size
MP4 (.mp4)	MPEG-4	Common web/video format, supports streaming
MOV (.mov)	QuickTime Movie	Developed by Apple, high quality
MKV (.mkv)	Matroska Video	Supports multiple audio and subtitle tracks
FLV (.flv)	Flash Video	Used for older web streaming
WMV (.wmv)	Windows Media Video	Compressed format for Windows

Obtaining Video Clips

Videos can be **created, recorded, or obtained** from various sources for use in multimedia projects.

Sources of Video:

1. **Recording:**
   - Using a camera, smartphone, or webcam to capture footage.
2. **Stock Video Libraries:**
   - Download from websites like Pexels, Pixabay, or Shutterstock.
3. **Screen Recording:**

- Capture on-screen actions using tools like OBS Studio or Camtasia.

4. **Existing Media:**
    - Use video segments from authorized or copyright-free content.

Considerations:

- Always check **copyright permissions**.
- Maintain **quality and resolution consistency** with other media.
- Trim or edit clips to fit the project purpose.

Shooting and Editing Video.

Creating professional-quality video requires careful **shooting** and **editing**.

A) Shooting Video

- **Plan the script or storyboard.**
- Choose suitable **camera angles and lighting**.
- Maintain **steady shots** using tripods.
- Capture **clear audio** with external microphones.
- Record **multiple takes** for better results.

B) Editing Video

Editing is done using software to organize, trim, and enhance video footage.

Editing Software:

- Adobe Premiere Pro
- DaVinci Resolve
- Final Cut Pro
- Filmora
- iMovie

Editing Tasks:

- Cutting unwanted parts
- Adding transitions and effects
- Synchronizing sound and subtitles
- Adjusting brightness, contrast, and color
- Exporting in the right format (MP4, MOV, etc.)

UNIT V

Making Multimedia:

- Creating a **multimedia project** is a structured process that involves combining **text, graphics, sound, video, and animation** into an interactive presentation or application.
- It requires **planning, creativity, technology, and teamwork**. The process goes through several stages and needs proper hardware, software, and professional coordination to ensure a successful output.

The Stage of Multimedia Project

A multimedia project is developed in **five main stages**:

a) Concept / Idea Development

- The process begins with an idea or concept.
- The objective of the project is identified — whether it is for education, training, entertainment, or business promotion.
- The target audience, goals, and expected outcomes are defined.

b) Planning and Design

- The project structure is planned using **storyboards, scripts, and flowcharts**.
- The layout, navigation system, and interaction style are designed.
- Resources such as graphics, sound, video, and animations are identified.

c) Production

- The production phase involves **creating and assembling multimedia elements** using specialized tools.
- Text, images, audio, video, and animations are integrated.
- Software tools like Adobe Photoshop, Premiere Pro, and Flash are used to develop the components.

d) Testing

- The project is tested for **technical accuracy, performance, and compatibility**.
- Errors such as broken links, sound distortions, or playback issues are identified and corrected.
- Feedback from users is collected and implemented.

e) Delivery / Distribution

- The final multimedia project is exported or packaged for distribution.

- It can be delivered via **CD/DVD, websites, mobile apps, or online streaming platforms**.
- The project must be optimized for performance on different devices.

The Intangible Needs

These are **non-physical or mental requirements** essential for multimedia development.

Key Intangible Needs:

- **Creativity:** To design engaging and visually appealing content.
- **Communication Skills:** To collaborate effectively among team members.
- **Project Management:** For proper scheduling, budgeting, and monitoring progress.
- **Team Coordination:** To ensure all departments work together smoothly.
- **Problem-Solving Skills:** To handle technical or design challenges efficiently.
- **Visualization and Innovation:** To bring new ideas and originality to the project.

These skills ensure that the project meets its goals and appeals to the target audience.

The Hardware Needs

Hardware forms the **physical foundation** of multimedia production. It includes input, processing, storage, and output devices.

Important Hardware Components:

a) Input Devices

Used to capture and enter data:

- Keyboard and Mouse
- Scanner
- Digital Camera / Webcam
- Microphone
- Graphic Tablet

b) Processing Devices

Used for rendering and editing multimedia:

- A powerful **computer** with a fast processor (Intel i7/i9, AMD Ryzen)
- High RAM capacity (minimum 16 GB recommended)
- High-performance **graphics card (GPU)** for animation and video rendering

c) Storage Devices

Used for saving large multimedia files:

- Hard Disk Drives (HDDs) or Solid State Drives (SSDs)
- External hard drives and USB devices
- Cloud storage for collaboration

d) Output Devices

Used to display or play back content:

- High-resolution **monitor**
- **Speakers** or headphones
- **Printers** for hard copies
- **Projectors** for presentations

The Software Needs

Software provides the **tools to create, edit, and integrate** multimedia components. Different types of software are used for specific purposes.

a) Text Editing Tools

- Microsoft Word, Notepad, Google Docs

b) Graphic Design Tools

- Adobe Photoshop, CorelDRAW, Illustrator

c) Audio Editing Tools

- Audacity, Adobe Audition, Sound Forge

d) Video Editing Tools

- Adobe Premiere Pro, Final Cut Pro, DaVinci Resolve

e) Animation Tools

- Adobe Animate, Blender, Maya, Toon Boom

f) Authoring Tools

- PowerPoint, Adobe Director, Authorware, Dreamweaver

Each software category contributes to a specific phase in multimedia production, from design to delivery.

An Authoring Systems Needs

An **authoring system** is a software tool used to **combine all multimedia elements** (text, sound, graphics, video, and animation) into one interactive program.

Functions of Authoring Systems:

- Importing and arranging multimedia elements
- Creating navigation and user interaction (buttons, menus)
- Adding scripting or logic for interactivity
- Previewing and testing the final output
- Exporting to web, CD, or executable format

Examples:

- Adobe Director
- Macromedia Author ware
- Microsoft PowerPoint

- Unity (for interactive multimedia and games)

Authoring tools act as the **final assembly platform** for all components of a multimedia project.

Multimedia Production Team

A multimedia project requires **teamwork and collaboration** among professionals with specialized skills.

Team Member	Role / Responsibility
Project Manager	Plans, organizes, and manages the overall project
Content Specialist	Provides subject matter or information for the project
Script Writer	Develops the storyline and narration
Graphic Designer	Designs images, illustrations, and visual layouts
Audio Specialist	Records, edits, and mixes sound and music
Video Specialist	Shoots and edits video clips
Animator	Creates animations and motion graphics
Programmer	Integrates multimedia elements and adds interactivity
Tester	Tests the final product for quality and functionality

Each member contributes a unique skill to ensure the project's success.

- ✓ Creating a multimedia project is a **team-driven and technology-based process**. It requires **planning, creativity, the right tools, and effective coordination** between team members.
- ✓ By using proper hardware, software, and authoring systems, multimedia professionals can produce **high-quality, interactive, and engaging content** for education, entertainment, and business applications.